

Basic Electrical Engineering

MODULE 5 — PART 1: COMPLETE NOTES

Module 5 (8 hrs): Measuring Instruments, Power Transmission & Distribution, Electrical Installations

For First-Year Students — written in simple, easy language

Course: Basic Electrical Engineering · Course Code: EE100

Module 5 — what it covers (as per the syllabus)

- Measuring instruments: classification, principles, essentials of an instrument
- Moving-coil (PMMC), moving-iron and dynamometer-type instruments; ammeters and voltmeters
- Measurement of power and energy
- Power transmission and distribution; low-voltage distribution (400 V / 230 V) through block diagrams
- Electrical installations: LT switchgear (SFU, MCB, ELCB, MCCB), types of wires and cables, earthing
- Types of batteries and their important characteristics

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1. Classification of Electrical Measuring Instruments

Electrical measuring instruments are broadly divided into two classes: **absolute instruments** and **secondary instruments**.

1.1 Absolute Instruments

Those instruments which give the value of the quantity to be measured in terms of the constants of the instrument and its deflection are called absolute instruments. A tangent galvanometer, used to measure current, is an example: it gives the current in terms of the tangent of the angle of deflection, the radius and number of turns of the coil, and the horizontal component of the earth's magnetic field. No previous calibration is necessary, and they are used only in standard laboratories.

1.2 Secondary Instruments

Those instruments in which the electrical quantity being measured is given directly by the deflection of the instrument are called secondary instruments. They are provided with a calibrated scale, the calibration being done with the help of an absolute or another calibrated instrument. These are the instruments most generally used in laboratories, power stations, substations and industries.

1.3 Types of Secondary Instruments (by function)

Type	Function	Examples
Indicating	Directly indicate the value of the quantity at the time of measurement.	Ammeter, voltmeter, wattmeter
Integrating	Measure the total quantity of electricity (Ah) or energy (Wh) over a time.	Ampere-hour meter, energy (watt-hour) meter
Recording	Give a continuous record of the variations of the quantity.	Recording voltmeter, recording ammeter

A recording instrument is simply an indicating instrument with a pen attached to its pointer, resting on a chart wrapped over a slowly moving drum.

2. Principles of Operation and Essentials of Indicating Instruments

An electrical instrument essentially consists of a movable element and a scale. The movement of the movable element is caused by using one or more of the following effects of current or voltage:

Effect	Type of instrument
Magnetic effect	Moving-iron instruments
Electrodynamic effect	Permanent-magnet moving coil (PMMC) and dynamometer type
Electromagnetic induction	Induction type instruments
Thermal effect	Hot-wire instruments
Chemical effect	Electrolytic instruments
Electrostatic effect	Electrostatic instruments

The three torques

For proper operation, an indicating instrument requires three torques:

- **Deflecting (operating) torque** — produced by the various effects of current or voltage; it moves the pointer from the zero position.
- **Controlling (restoring) torque** — provided by a spring or by gravity; it opposes the deflecting torque so that the pointer comes to rest at a position where the two torques are equal.
- **Damping torque** — provided by air friction or eddy currents; it brings the pointer to its final position without oscillation, allowing quick and accurate readings.

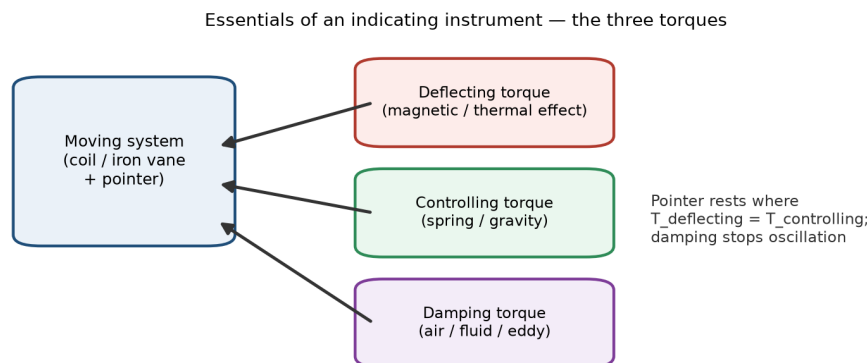


Figure 1: The three torques of an indicating instrument.

3. Controlling Torque and Damping Torque

3.1 Spring Control

A spiral hairspring of a non-magnetic material such as phosphor bronze is attached to the moving system. As the pointer deflects, the spring twists in the opposite direction and provides the controlling torque. Since the torsion of a spiral spring is proportional to the angle of twist:

$$T_C \propto \theta$$

The pointer rests where $T_d = T_C$. In an instrument with a uniform deflecting torque (e.g. PMMC, where $T_d \propto I$), spring control gives a **linear, evenly-spaced scale**. Its advantages are that levelling is not required if the parts are balanced, springs also serve as current leads, and there is little extra weight; its disadvantages are the effect of temperature, difficulty of adjustment and possible damage by accidental stress.

3.2 Gravity Control

A small adjustable weight W is attached to the moving system. In the zero position the weight hangs vertically and gives no torque; when the pointer deflects, the weight moves from the vertical and provides the controlling torque. Only the component $W \sin \theta$ acts, so:

$$T_C = W l \sin \theta \quad \text{so} \quad T_C \propto \sin \theta$$

With gravity control the scale is **non-uniform**. Its advantages are lower cost, freedom from temperature effects and fatigue, and easy adjustment; its disadvantages are that the instrument must be kept vertical, the weight increases the moving-system weight, and the scale is non-uniform.

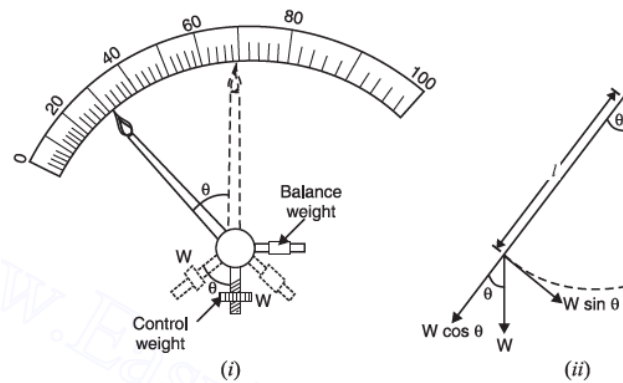


Figure 2: Gravity control of the moving system.

3.3 Damping Torque

Damping brings the pointer smoothly to its final reading. Without it the instrument may be under-damped (oscillating) or over-damped. The three methods are:

- **Air friction damping** — a light vane moves in a closed air chamber.
- **Fluid friction damping** — a vane moves in a pot of oil.
- **Eddy-current damping** — a metal disc moving in a magnetic field has eddy currents induced in it, which oppose the motion.

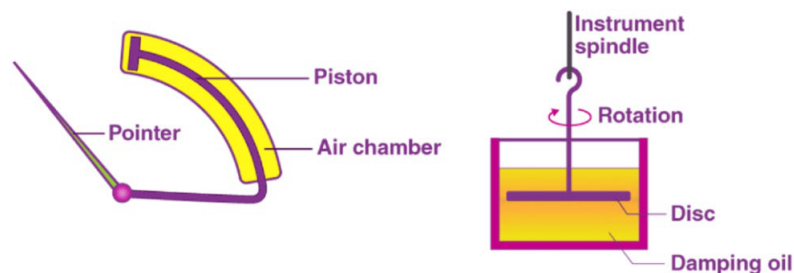


Figure 3: Damping methods used in indicating instruments.

4. Ammeters and Voltmeters

An **ammeter** measures the current in a circuit. It is connected in **series** with the circuit, so that the current to be measured (or a fraction of it) passes through it. It must carry this current without injury and without abnormally increasing the circuit resistance, so an ammeter is designed to have a **low resistance**.

A **voltmeter** measures the potential difference between two points of a circuit. It is connected in **parallel** with the circuit or part of it. It must have enough resistance so that it is not injured by the current through it and does not materially affect the circuit current, so a voltmeter is designed to have a **high resistance**.

The basic principle of the ammeter and voltmeter is the same — both are current-operated devices, the deflecting torque being produced by a current through their operating coils. Hence the same instrument can serve as either, with proper design.

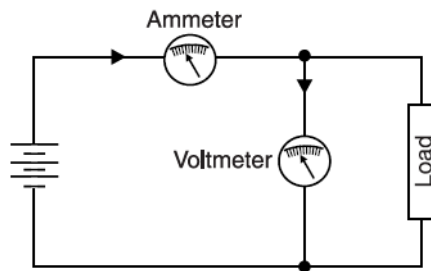


Figure 4: Ammeter in series and voltmeter in parallel.

5. Permanent-Magnet Moving Coil (PMMC) Instruments

These instruments are used as ammeters or voltmeters and are suitable for **DC only**. They are based on the principle that when a current-carrying conductor is placed in a magnetic field, a mechanical force acts on it. The coil carrying the operating current is placed in the field of a permanent magnet and is attached to the moving system, which carries the pointer.

5.1 Deflecting Torque

Because of the soft-iron core, the field in the air-gap is radial, so the coil sides always move at right angles to the field. If B is the flux density, l the length, b the breadth, N the number of turns and I the current, the force on each side is $F = B I l$, and:

$$T_d = B I N A \text{ newton-metre } (A = b \times l)$$

Since B , N and A are fixed, $T_d \propto I$. With spring control $T_c \propto \theta$, and at rest $T_d = T_c$, so $\theta \propto I$. Hence PMMC instruments have a **uniform (linear) scale**.

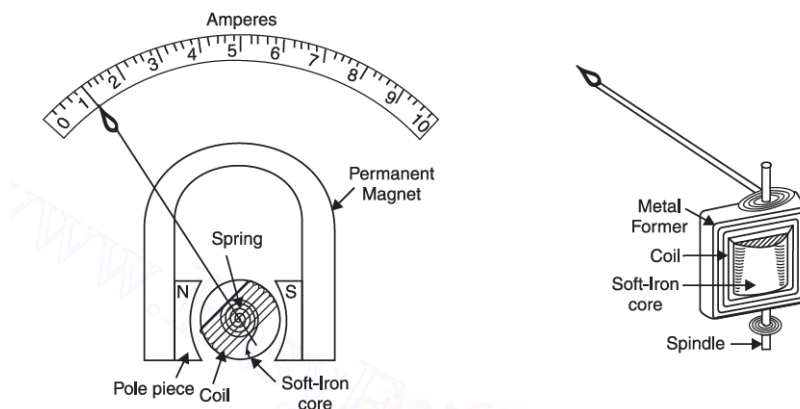


Figure 5: Construction of a PMMC instrument.

5.2 Working and Why DC Only

When the operating current flows through the coil, a mechanical torque acts on it and the pointer moves over the scale. If the current is reversed, the deflecting torque also reverses (the permanent-magnet field being unchanged), so the pointer would deflect below zero — this is prevented by a spring stop. Since the torque reverses with the current, the instrument can measure direct current and voltage only.

5.3 Advantages, Disadvantages and Applications

Advantages	Disadvantages
Uniform scale; very accurate and reliable.	Cannot be used for AC measurements.
Very effective eddy-current damping (aluminium former in a strong field).	About 50% costlier than moving-iron instruments.
High efficiency, requires very little power.	Errors due to ageing of the magnet or the control springs.
No hysteresis loss (constant flux); stray fields have little effect.	

Applications: measurement of direct currents and voltages, in DC galvanometers to detect small currents, and in ballistic galvanometers for measuring changes of magnetic flux linkages.

6. Moving-Iron (MI) Instruments

These are principally used for the measurement of alternating currents and voltages, though they can also be used for DC. There are two types:

- **Attraction type** — a single soft-iron vane mounted on the spindle is attracted towards the coil when the operating current flows through it.
- **Repulsion type** — two soft-iron vanes are used, one fixed with the coil and the other movable on the spindle.

6.1 Attraction Type

It consists of a fixed cylindrical coil (solenoid) and an oval soft-iron piece attached to the spindle so that it can move in and out of the coil. A pointer on the spindle deflects with the motion of the iron. The controlling torque is provided by a spiral spring at the top; the springs do not carry current (the current is carried by the stationary coil). Damping is by an aluminium vane moving in a closed chamber.

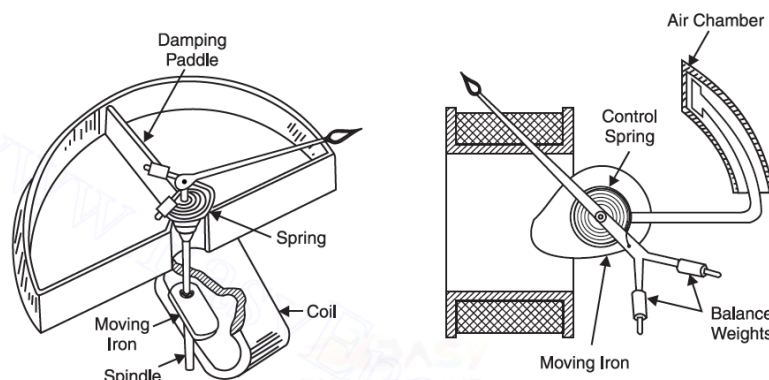


Figure 6: Attraction-type moving-iron instrument.

6.2 Repulsion Type

Two soft-iron vanes are surrounded by a fixed hollow coil carrying the operating current. One vane is fixed (wedge-shaped, larger radius) and the other is movable and mounted on the spindle with the pointer. When current flows, both vanes are magnetised with similar polarity at the same ends, so they **repel** each other and the movable vane deflects the pointer. Reversing the current reverses the magnetisation of both vanes, so the direction of the deflecting torque is unchanged — hence these instruments can be used for both DC and AC.

The deflecting torque is proportional to the product of the pole strengths, which is proportional to the square of the coil current:

$$T_d \propto I^2 \quad \text{so} \quad \theta \propto I^2$$

Thus the scale of a moving-iron instrument is **non-uniform** — crowded at the beginning and spread out near the end. The non-linearity can be partly corrected by the shaping and positioning of the vanes.

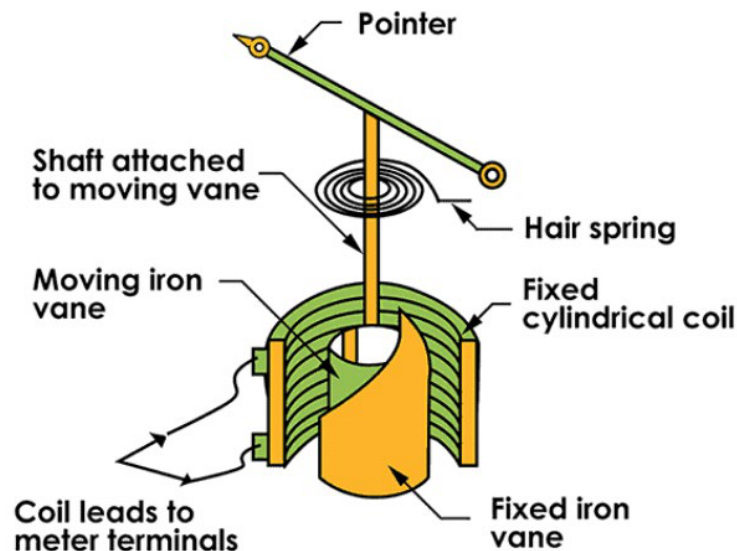


Figure 7: Repulsion-type moving-iron instrument.

6.3 Errors and Characteristics

Errors with both DC and AC: hysteresis (readings higher when current increases than when it decreases — reduced by using Mu-metal vanes at low flux density), stray fields (reduced by an iron case), temperature (affects resistance and spring stiffness) and friction (reduced by a high torque-to-weight ratio). **Error with AC only:** a change of frequency changes the coil impedance; this can be corrected by a capacitor $C = L_m/R_s^2$ in parallel with the multiplier.

Advantages	Disadvantages
Less expensive, robust and simple.	Non-linear scale.
Can be used for both DC and AC.	Not as sensitive as PMMC instruments.
High operating torque and reasonably accurate.	Frequency error in AC measurements.

Moving-iron instruments are primarily used for AC measurements; they are not used for DC because their characteristics are inferior to those of PMMC instruments.

7. Wattmeter and Measurement of Energy

A wattmeter measures the electric power given to or developed by a circuit. In a DC circuit power can be found from voltmeter and ammeter readings ($P = VI$), but in an AC circuit power also depends on the phase shift between voltage and current ($P = VI \cos \phi$), so a wattmeter is necessary. The two principal types are the **dynamometer wattmeter** (for both DC and AC) and the **induction wattmeter** (for AC only).

7.1 Dynamometer Wattmeter

It works on the dynamometer principle — a mechanical force exists between two current-carrying coils. It has a fixed current coil (connected in series with the load) and a movable potential coil (connected in parallel through a high resistance). Its deflection is proportional to $V I \cos \phi$, so it reads the real power directly.

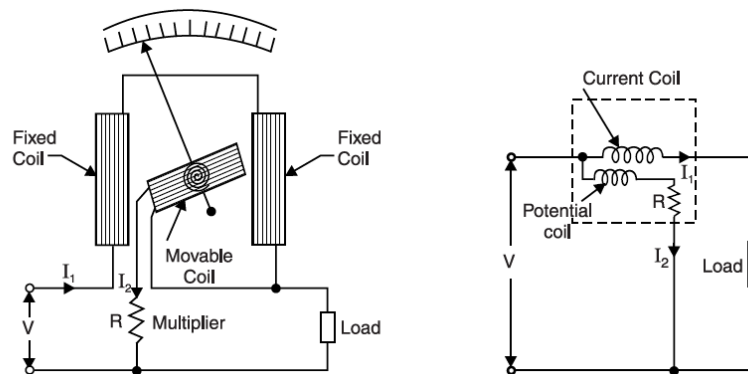


Figure 8: Dynamometer wattmeter connection.

Advantages	Disadvantages
Can be used for both AC and DC power.	At low power factors, the potential-coil inductance causes serious errors.
Uniform scale (unlike dynamometer ammeters/voltmeters).	Readings may be affected by stray magnetic fields (reduced by a soft-iron case).
High accuracy by careful design.	The coils may burn out if current or voltage limits are exceeded.

Wattmeter ratings: a wattmeter has a current-coil circuit and a potential-coil circuit, and both ratings are marked. At low power factor the current through the current coil may exceed the safe value while the reading is well below full scale, so an ammeter and voltmeter should be used along with a wattmeter when the load power factor is low or unknown.

7.2 Measurement of Energy

Electrical energy is measured by an **energy meter** (watt-hour meter), which is an integrating instrument. It records the total energy consumed in kilowatt-hours (kWh, commonly called a "unit"). One unit = 1 kWh = the energy used by a 1 kW load for 1 hour.

8. Power Transmission and Distribution — Structure

An electric power system is a network of electrical components that supply, transfer and use electric power. It has three main parts:

- **Generation** — the source of energy.
- **Transmission** — the transfer of energy over long distances (close to the speed of light).
- **Distribution** — the consumption of energy.

Electrical power transmission involves the bulk movement of energy from a generating station to a substation, where the voltage is transformed and then distributed to consumers or other substations. The interconnected lines that enable this movement are called the transmission network, or the **power grid**.

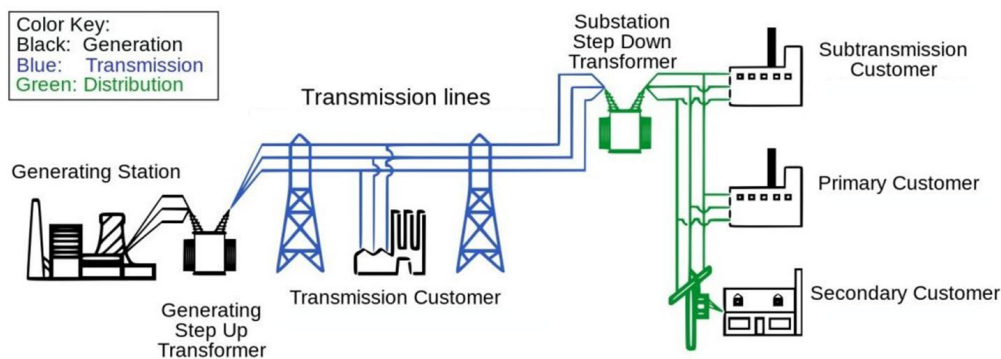


Figure 9: General structure of an electric power system.

9. Generation, Transmission and Distribution

9.1 Generation

In the generating station the fuel (coal, water, nuclear energy, etc.) is converted into electrical energy. Power is generated in the range of **11 kV to 25 kV**, which is then stepped up for long-distance transmission. Generating plants are mainly of three types — thermal, hydro and nuclear. The generator converts mechanical energy into electrical energy, and the transformer transfers power with very high efficiency from one voltage level to another.

9.2 Primary Transmission

Before being sent over the transmission lines, the generated voltage (11-33 kV) is stepped up to anywhere between **100 kV and 700 kV or more**, depending on the distance — the longer the distance, the higher the voltage. The reason is to reduce the **I^2R loss**: when the voltage is stepped up, the current falls for the same power, and since the loss is I^2R , it is greatly reduced. This stage — the transfer of a large quantity of power from the generating station to a substation via overhead lines — is called primary transmission.

9.3 Secondary Transmission

At the receiving station the voltage is stepped down to typically **33-66 kV** and sent to substations nearer the load centres (cities, villages and urban areas). At the substation it is stepped down once more, usually to about **11 kV**. From here the transmission phase gives way to the distribution phase.

9.4 Distribution System

The part of the power system that distributes electric power for local use is the distribution system — the electrical system between the substation (fed by the transmission system) and the consumers' meters. It consists of:

Part	Description
Feeders	A conductor that connects the substation to the area where power is to be distributed. Generally no tappings are taken, so the current stays the same throughout; the design consideration is the current-carrying capacity.
Distributors	A conductor from which tappings are taken for supply to the consumers.
Service mains	A small cable that connects the distributor to the consumer's terminals.

10. Low-Voltage Distribution (400 V / 230 V)

The final stage of the system supplies power at low voltage through the distribution transformer. The usual low-voltage (LT) supply is:

- **400 V, three-phase (3-phase, 4-wire)** — for three-phase loads such as motors in small-scale industries and commercial buildings.
- **230 V, single-phase** — between one phase and neutral, for domestic lighting, fans and small appliances.

Structure of the power system and the low-voltage distribution

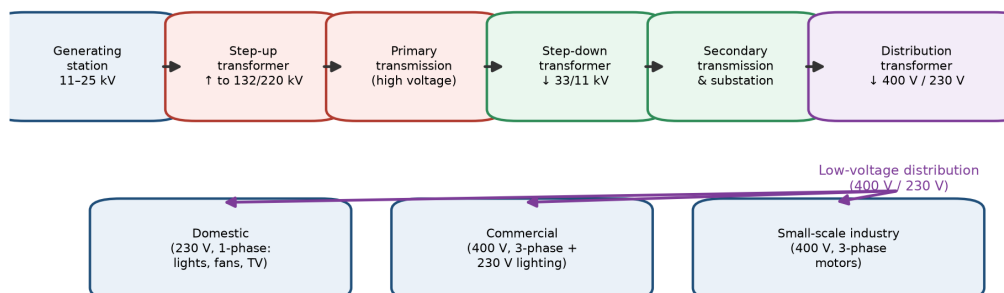


Figure 10: Low-voltage distribution (400 V / 230 V) to domestic, commercial and small-scale industry consumers.

The block diagram shows the path: generating station → step-up transformer → primary transmission → step-down transformer → secondary transmission and substation → distribution transformer (400 V / 230 V) → domestic, commercial and small-scale industry consumers.

11. Components of LT Switchgear

The apparatus used for switching, controlling and protecting electrical circuits and equipment is called **switchgear**. The main low-tension (LT) switchgear components are the switch fuse unit (SFU), the miniature circuit breaker (MCB), the moulded-case circuit breaker (MCCB) and the earth-leakage circuit breaker (ELCB).

11.1 Switch Fuse Unit (SFU)

A combination of a switch and a fuse is called an SFU. The **switch** isolates the circuit from the supply for repair and maintenance and is manually operated. The **fuse** is a protective device that acts quickly under abnormal conditions to disconnect the circuit; it works on the heating effect of current, is made of a thin strip or wire of copper, zinc, aluminium or silver, and is always connected in series. When too much current flows, the fuse element softens and opens the circuit, protecting the wiring and people.



Switch fuse unit

Figure 11: Switch fuse unit (SFU).

11.2 Miniature Circuit Breaker (MCB)

An MCB is an electromechanical device that makes and breaks the circuit in normal operation and disconnects it when the current exceeds a preset value during a fault. It has a high fault capacity, is current-limiting and trip-free, and uses thermal and magnetic operation. It protects against **overload and short circuit**. MCBs are rated for 240 V single-phase, 415 V three-phase and 220 V DC, with current ratings from 0.5 A to 63 A.



Figure 12: Miniature circuit breaker (MCB).

11.3 Moulded-Case Circuit Breaker (MCCB)

An MCCB is similar to an MCB but is used for circuits with currents from **63 A to 3000 A**. Its working is based on a thermal mechanism: a bimetallic contact expands and contracts with temperature changes. Under normal conditions the contacts are closed; under overload or short circuit the current exceeds its safe value, heat is generated, and the contacts open to interrupt the circuit. Because interrupting a high current forms an arc, MCCBs have **arc extinguishers**. They offer adjustable trip settings and are used in industrial, commercial and residential installations, usually in the distribution panel.



Figure 13: Moulded-case circuit breaker (MCCB).

11.4 Earth-Leakage Circuit Breaker (ELCB)

An ELCB is a safety device used in installations with high earth impedance to prevent shock. It detects small stray voltages on the metal enclosures of equipment (or a difference in current between the line and neutral wires) and interrupts the circuit if a dangerous voltage or leakage current is detected. It protects personal safety against electric shock and also guards against short circuit and overload. Modern installations often use residual-current devices (RCDs/RCCBs) which detect leakage current directly.



Figure 14: Earth-leakage circuit breaker (ELCB).

12. Types of Wires and Cables

A wire is a single conductor, while a cable is a group of wires insulated from one another and enclosed in a protective covering. Common types of wire are:

Type of wire	Description and use
V.I.R. (vulcanised Indian rubber)	Rubber-insulated wire; cheap, used in dry places and for low-voltage internal wiring.
PVC (polyvinyl chloride)	Most common; tough, moisture-proof and fire-retardant; used for domestic and industrial wiring.
CTS / TRS (cabinet/tough rubber sheathed)	Rubber-sheathed wire, moisture- and chemical-proof; suitable for damp climates but not outdoors.
Lead / metal sheathed	Wire with a lead-aluminium alloy sheath that protects against dampness and mechanical damage; the sheath is earthed; costly, used for low-voltage installations.

Depending on how the wires are fixed, the usual wiring systems are **cleat wiring**, **batten (CTS/TRS) wiring**, **metal (lead) sheathed wiring**, **casing-and-capping** and **conduit wiring**. Cables are classified as single-core or multi-core, and as armoured (with a steel tape/wire covering for mechanical protection, used underground) or unarmoured.

13. Earthing

Earthing (or grounding) is the process in which the instantaneous discharge of electrical energy takes place by transferring charges directly to the earth through a low-resistance wire. It is the connection of electrical machinery to the general mass of the earth with a conducting material of very low resistance.

Earthing brings the equipment to **zero (earth) potential** and avoids shock to the operator under fault conditions. The non-current-carrying metallic parts of the equipment are connected to an earth electrode or earthing mat placed near or below the ground level.

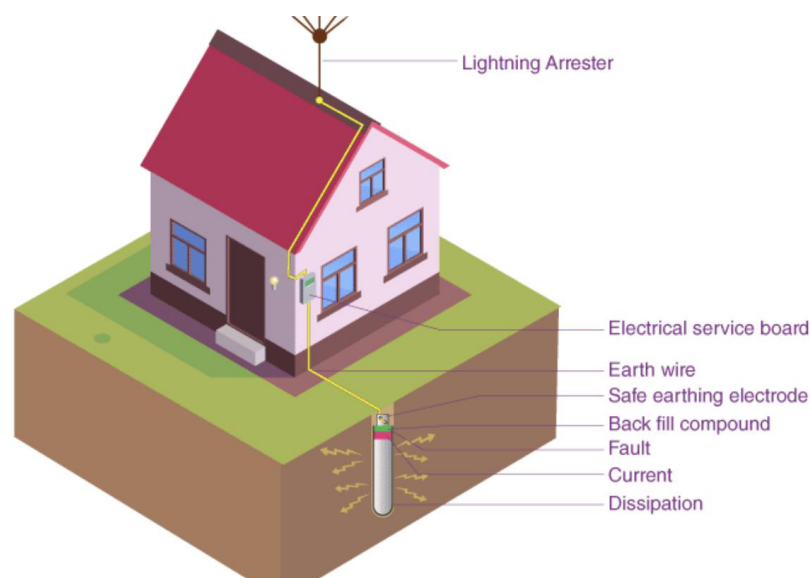


Figure 15: Earthing of electrical equipment.

Types of Earthing

Type	Description
Pipe earthing	A galvanised steel pipe is buried vertically into the ground. Affordable and low-maintenance; suitable for areas of high soil resistivity.
Plate earthing	A metal plate (usually copper or galvanised iron) is buried vertically. Commonly used in residential and small commercial buildings.
Strip (wire) earthing	Strip electrodes are buried in horizontal trenches; preferred for rocky terrain.
Rod earthing	A metal rod (copper or galvanised iron) is driven vertically into the ground; effective where soil resistivity is low.

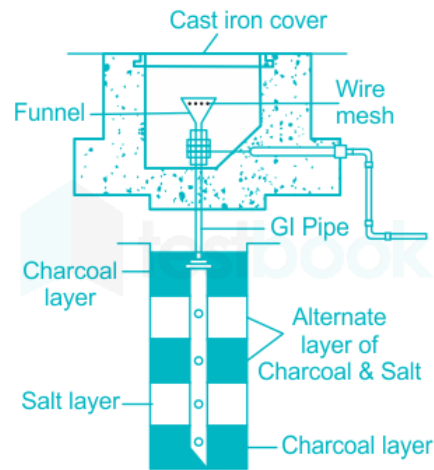


Figure 16: Pipe earthing.

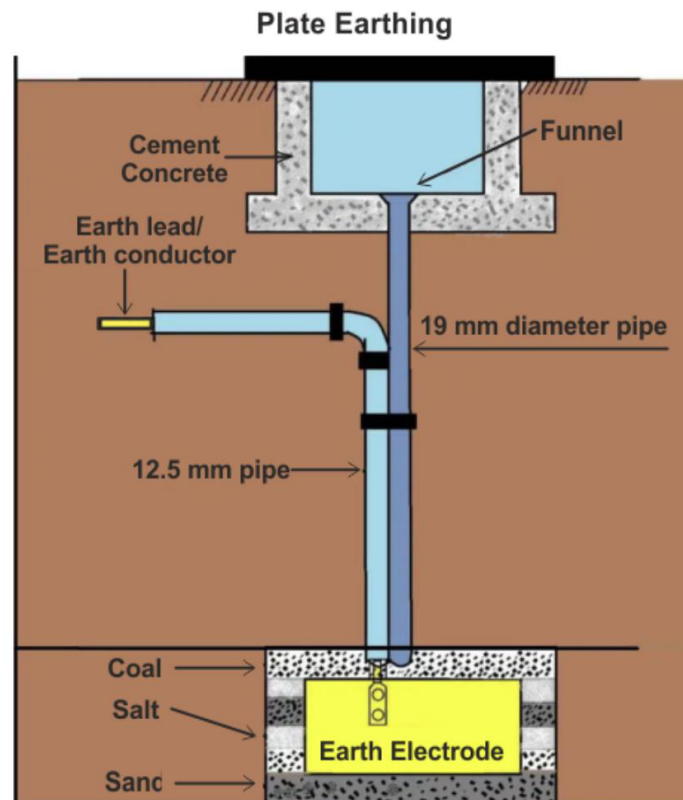


Figure 17: Plate earthing.

14. Batteries

A battery is a device that stores chemical energy and converts it into electrical energy. It consists of one or more electrochemical cells. Batteries are of two kinds:

Class	Description	Examples
Primary cells	Non-rechargeable; the chemical reaction is not reversible, so the cell is discarded after use.	Dry cell (Leclanché), alkaline cell
Secondary cells	Rechargeable; the reaction is reversible, so the cell can be charged and used again.	Lead-acid, nickel-cadmium (Ni-Cd), lithium-ion

Important Characteristics of Batteries

- **Capacity** — the quantity of charge a battery can deliver, in ampere-hours (Ah); it equals the current $\times$ the time of discharge.
- **Voltage** — the emf of the battery; a lead-acid cell gives about 2 V, so a 12 V battery has six cells.
- **Energy density** — energy stored per unit weight or volume; lithium-ion is the highest.
- **Internal resistance** — should be small; it causes a voltage drop when current flows.
- **Cycle life** — the number of charge-discharge cycles a secondary cell can give before its capacity falls.
- **Self-discharge** — the loss of charge when the battery is not in use.
- **Efficiency** — the ratio of energy obtained on discharge to the energy supplied on charge.

- **Specific gravity** — for a lead-acid cell, the state of charge is checked from the specific gravity of the electrolyte (about 1.28 when fully charged).

15. Solved Numerical Examples

Example 1: Shunt for an ammeter

Problem: a PMMC meter of resistance $100\ \Omega$ gives full-scale deflection for $10\ \text{mA}$. Find the shunt resistance to use it as a $1\ \text{A}$ ammeter.

$$R_{sh} = R_m I_m / (I - I_m) = 100 \times 0.01 / (1 - 0.01) = 1.01\ \Omega$$

Example 2: Multiplier for a voltmeter

Problem: the same meter ($100\ \Omega$, $10\ \text{mA}$) is to be used as a $100\ \text{V}$ voltmeter. Find the series multiplier.

$$R_s = V / I_m - R_m = 100 / 0.01 - 100 = 9900\ \Omega$$

Example 3: Energy consumed

Problem: a load of $1.5\ \text{kW}$ works for $8\ \text{hours}$. Find the energy consumed in kWh and the cost at ₹5 per unit.

$$\text{Energy} = 1.5 \times 8 = 12\ \text{kWh}; \text{cost} = 12 \times 5 = ₹60$$

Example 4: Battery capacity

Problem: a battery delivers a steady current of $5\ \text{A}$ for $4\ \text{hours}$. Find its capacity.

$$\text{Capacity} = I \times t = 5 \times 4 = 20\ \text{Ah}$$

16. Quick Reference — Key Points and Formulae

Item	Key point / formula
Absolute / secondary instruments	Absolute gives the value from the instrument constants; secondary reads directly from a calibrated scale.
Indicating / integrating / recording	Reads at the instant / totals over time (Ah, Wh) / gives a continuous record.
Three torques	Deflecting, controlling and damping torques are all needed.
Spring / gravity control	$T_C \propto \theta$ (linear scale) / $T_C \propto \sin \theta$ (non-uniform scale).
Damping methods	Air friction, fluid friction and eddy-current damping.
Ammeter / voltmeter	Ammeter in series, low resistance / voltmeter in parallel, high resistance.
PMMC	DC only; $T_d = BINA \propto I$; uniform scale.
Moving iron	AC and DC; $T_d \propto I^2$; non-uniform scale.
Wattmeter	Reads $P = VI \cos \phi$; dynamometer type for DC and AC.
Energy meter	Integrating; measures energy in kWh (1 unit = 1 kWh).
Power system	Generation → transmission → distribution.
Transmission voltages	Generated 11–25 kV; primary transmission 100–700 kV; secondary 33–66 kV, then 11 kV.
LV supply	400 V three-phase and 230 V single-phase.
Distribution parts	Feeders, distributors and service mains.
LT switchgear	SFU, MCB (0.5–63 A), MCCB (63–3000 A), ELCB.
Earthing	Connects metal parts to earth at zero potential; pipe, plate, strip and rod earthing.
Batteries	Primary (non-rechargeable) and secondary (rechargeable); capacity in Ah.